HS

Harper Strickland

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Curriculum Vitae

RESEARCH INTERESTS

- Speech Processing
- Audio Data & Sound Synthesis
- Ethical Data Science
- Assistive Technology

DEGREES

MAS – Data Science and Engineering – June 2025 University of California San Diego

Engineering Honor Societies: IEEE Eta Kappa Nu (HKN), Tau Beta Pi

Topics include:

Computational Speech Processing, Bioacoustic Classification Models, Machine Learning, Neural Networks, Probability and Statistics, Python for Data Analysis, Data Management Systems, Scalable Data Analysis, Data Visualization, Data Integration and ETL

Capstone Project:

Avian Language Models: Rethinking Bioacoustic Pipelines with Self-Supervised Learning. Ablation studies of bioacoustic model performance: encoder, Q-Former, large language model (LLM). Hari Prakash, **Harper Strickland**, Xinyu Bao, Jiawei Li, Alex Lehman, Sean Perry, Ryan Kastner. UC San Diego Library Digital Collections, June 2025. https://doi.org/10.6075/J0NV9JM2

MBA – Business Administration University of Phoenix, Arizona

Topics include:

Statistics and Research Methods, Information Systems, Project Management

BA – Cognitive Science, Music, Computer Science University of Virginia

Topics include:

Linguistics, Symbolic Logic, Computer Sound Generation, Human Computer Interface, Invention and Design, Cognitive Psychology, Discrete Math, Digital Logic Design, Database Systems

PEER-REVIEWED PUBLICATIONS

AUTALIC: A Dataset for Anti-AUTistic Ableist Language In Context. First annotated dataset for detecting anti-autistic ableist language, with machine learning analysis. Naba Rizvi, **Harper Strickland**, Daniel Gitelman, Alexis Morales Flores, Tristan Cooper, Aekta Kallepalli, Akshat Alurkar, Haaset Owens, Saleha Ahmedi, Isha Khirwadkar, Imani Munyaka, Nedjma Ousidhoum.
Proceedings of the Association for Computational Linguistics (ACL), July 2025. https://doi.org/10.18653/v1/2025.acl-long.1022

'I Hadn't Thought About That': Creators of Human-like AI Weigh in on Ethics & Neurodivergence. Analysis of interviews with 16 developers about accessibility, neurodiversity, and ethical implications. Naba Rizvi, Taggert Smith, Tanvi Vidyala, Mya Bolds, **Harper Strickland**, Andrew Begel, Rua Williams, Imani Munyaka.
Association for Computing Machinery (ACM) Conference on Fairness, Accountability, and Transparency (FAccT), June 2025. https://doi.org/10.1145/3715275.3732218

PUBLIC DATASETS

LJ2 Corpus: 1 voice, 26,200 transcribed sound files, 48 hours of speech audio
Triples the size of the popular LJ Speech Dataset, with over 8 times more non-fiction source texts, employing downsampling to increase alignment with modern word usage frequency. It can be used alone in 12-, 24-, or 48-hour segments, and with any model architecture designed to work with LJ Speech.
Documentation and Repository: https://github.com/speakingofdata/LJ2_Corpus

80 Excerpts: 4 voices, 80 transcriptions, 320 sound files
60 text selections from LJ Speech Dataset or LJ2 Corpus (non-fiction), and 20 from other sources (fiction). Recordings of four adult Americans reading each excerpt, 80 wav files for each voice.
Documentation and Repository: https://github.com/speakingofdata/80_Excerpts

ADDITIONAL EDUCATION

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS) October 2025
Research on design, evaluation, use, and education related to computing for people with disabilities.

Neurodiversity Affirming AAC – Northwest Augmentative Communication Society September 2025
Conference featuring AAC research and first-hand perspectives, prioritizing user-defined objectives.

North American Summer School for Logic, Language, and Information June 2025
Topics include: Semantics, Generalized Quantifiers, Computational Social Science on Linguistic Data

WORK EXPERIENCE

Industrial Engineering Supervisor – UPS 8 years
Managed ongoing optimization of complex logistics, warehouse, and distribution systems. Analyzed operational data, recommended and implemented process improvements to achieve production goals. Maintained district service area map data and on-road work measurements for 4 states.
Select Accomplishments:

- Forecasted staffing, equipment, and volume projections for largest package delivery facility in USA.
- Oversaw transition to first automated warehouse facility in Colorado. Steered project management after loss of institutional knowledge caused over 200 overdue transition tasks. Guided engineering plans to achieve successful opening of operations on time while meeting safety, performance, and cost targets.

Community Development Consultant – U.S. Peace Corps 2 years
Opened a Community Education Center in Ukraine, hired and trained director and staff, developed a sustainable business plan, and created programs serving 270 local participants of all ages. Conducted professional training in multilingual and cross-cultural environments.
Select Accomplishments:

- Successfully secured and administered USAID-funded grant supporting NGO enterprise.
- Founded and chaired Special Needs Working Group for Peace Corps Volunteers.
- Trained and mentored center director, who subsequently envisioned and implemented educational projects independently, demonstrating capacity to maintain center programming without external assistance.

Test Scorer – Pearson 4 years
Graded science responses at multiple grade levels for seasonal standardized tests.

Teaching Assistant – University of Virginia 2 years
Facilitated laboratory participation and assessed performance on tests, quizzes, and assignments for introductory Computer Science C++ programming course, 50 students per semester.

SKILLS

- Speech Processing: pitch/formant extraction, corpus development, voice transmission codecs, text-to-speech (TTS), automatic speech recognition (ASR), Praat, Kaldi, ESPnet, SpeechBrain, Unsloth
- Data Management and Analysis: Python, Pandas, SciPy
- Machine Learning, LLMs, Deep Learning, Neural Networks: Scikit-learn, TensorFlow, PyTorch
- Mathematical Basis for Data Models: probability, statistics, principle component analysis (PCA)
- Data Visualization: Tableau, Matplotlib, Seaborn, Plotly, Dash
- Natural Language Processing (NLP): NLTK, SpaCy
- Relational Database Systems: SQL, Postgres, Oracle
- Graph Database Systems: XML, XQuery, JSON, PartiQL, Neo4j, Cypher, SparQL
- Scalable Data Analysis, Cloud-based and Parallel Computing: Hadoop, Spark, AWS
- Time Series Analysis, Regression Models, Supervised and Unsupervised Learning, ETL, Knowledge Graph
- Additional Programming Experience: C++, R, Perl, LaTeX, RTcmix
- Spoken languages: English (native), Ukrainian (proficient), Russian (basic)